

Task 1 – Accident Prediction

Dataset 1

Weather	Road	Traffic	Engine problem	Accident
Rain	bad	high	no	yes
Snow	average	normal	yes	yes
Clear	bad	light	no	no
Clear	good	light	yes	yes
Snow	good	normal	no	no
Rain	average	light	no	no
Rain	good	normal	no	no
Snow	bad	high	no	yes
Clear	good	high	yes	no
Clear	bad	high	yes	yes

Target = **Accident (yes/no)**

There are 5 “Yes” and 5 “No”.

→ $P(\text{Yes})=0.5$ $P(\text{No})=0.5$

We need $P(\text{Accident} = \text{Yes} \mid \text{Weather}=\text{Rain}, \text{Road}=\text{Good}, \text{Traffic}=\text{Normal}, \text{EngineProblem}=\text{No})$.

Step 1 – Conditional probabilities

Feature	Value	$P(\text{value} \mid \text{Yes})$	$P(\text{value} \mid \text{No})$
Weather	Rain	$1/5 = 0.2$	$2/5 = 0.4$
Road	Good	$1/5 = 0.2$	$2/5 = 0.4$
Traffic	Normal	$2/5 = 0.4$	$2/5 = 0.4$
Engine Problem	No	$3/5 = 0.6$	$4/5 = 0.8$

Step 2 – Posterior probabilities

$P(\text{Yes} \mid X) \propto 0.5(0.2)(0.2)(0.4)(0.6) = 0.0048$

$P(\text{No} \mid X) \propto 0.5(0.4)(0.4)(0.4)(0.8) = 0.0256$

Prediction: No (accident unlikely).

Task 2 – Weather-Based Game Prediction

Dataset 2 (“Play Tennis” classic data)

Target = **Play (yes/no)**

- Yes = 9, No = 5 → $P(\text{Yes})=9/14 \approx 0.64$ $P(\text{No})=0.36$

Q 1 : (sunny, hot, high, false)

Feature	Value	$P(\text{value} \text{Yes})$	$P(\text{value} \text{No})$
Outlook	Sunny	$2/9=0.22$	$3/5=0.6$
Temperature	Hot	$2/9=0.22$	$2/5=0.4$
Humidity	High	$3/9=0.33$	$4/5=0.8$
Windy	False	$6/9=0.67$	$2/5=0.4$

Compute:

$$P(\text{Yes} | X) \propto 0.64(0.22)(0.22)(0.33)(0.67) = 0.0069$$

$$P(\text{No} | X) \propto 0.36(0.6)(0.4)(0.8)(0.4) = 0.0276$$

Prediction: No (do not play). (Table agrees.)

Q 2 : (overcast, cool, high, true)

Feature		$P(\text{value} \text{Yes})$	$P(\text{value} \text{No})$
Outlook	Overcast	$4/9=0.44$	$0/5=0$
Temperature	Cool	$3/9=0.33$	$1/5=0.2$
Humidity	High	$3/9=0.33$	$4/5=0.8$
Windy	True	$3/9=0.33$	$3/5=0.6$

$$P(\text{Yes} | X) \propto 0.64(0.44)(0.33)(0.33)(0.33) = 0.0103$$

$$P(\text{No} | X) \propto 0.36(0)(0.2)(0.8)(0.6) = 0$$

Prediction: Yes (play). (Consistent with table.)

Task 3 – Loan Approval Prediction

Dataset 3 (5 rows)

Employment	Credit History	Income	LoanApproved
Employed	Good High	Yes	
Unemployed	Bad Low	No	
Employed	Good Medium	Yes	
Employed	Bad Medium	No	
Unemployed	Good Low	Yes	

Target = LoanApproved.

$P(\text{Yes}) = 3/5 = 0.6$ $P(\text{No}) = 2/5 = 0.4$

Q 1 : Employed, Good, Medium

Feature	P(value Yes)	P(value No)
Employment=Employed	2/3	1/2
Credit=Good	3/3	0
Income=Medium	1/3	1/2

$P(\text{Yes}|X) \propto 0.6(0.67)(1)(0.33) = 0.132$

$P(\text{No}|X) \propto 0.4(0.5)(0)(0.5) = 0$

Prediction: Yes (approved).

Q 2 : Unemployed, Bad, Low

Feature	P(value Yes)	P(value No)
Employment=Unemployed	1/3	1/2
Credit=Bad	0	1
Income=Low	1/3	1/2

$P(\text{Yes}|X) \propto 0.6(0.33)(0)(0.33) = 0$

$P(\text{No}|X) \propto 0.4(0.5)(1)(0.5) = 0.1$

Prediction: No (not approved).

Task 4 – Disease Diagnosis

Dataset 4 (5 rows)

Fever	Cough	Fatigue	Travel History	Diagnosis
Yes	Yes	Yes	Yes	Positive
No	Yes	No	No	Negative
Yes	No	Yes	No	Positive
No	Yes	No	Yes	Negative
Yes	No	Yes	No	Positive

Target = Diagnosis

$P(\text{Pos}) = 3/5 = 0.6$ $P(\text{Neg}) = 0.4$



Q 1 : Fever = Yes, Cough = No, Fatigue = Yes, Travel = No

Feature	$P(\text{value} \text{Pos})$	$P(\text{value} \text{Neg})$
Fever=Yes	$3/3 = 1.0$	$0/2 = 0$
Cough=No	$2/3 = 0.67$	$0/2 = 0$
Fatigue=Yes	$3/3 = 1.0$	0
Travel=No	$2/3 = 0.67$	$1/2 = 0.5$

$P(\text{Pos} | X) \propto 0.6(1)(0.67)(1)(0.67) = 0.27$

$P(\text{Neg} | X) \propto 0.4(0)(0)(0)(0.5) = 0$

Prediction: Positive.

Q 2 : Fever = No, Cough = Yes, Fatigue = No, Travel = No

Feature	$P(\text{value} \text{Pos})$	$P(\text{value} \text{Neg})$
Fever=No	0	1
Cough=Yes	$1/3$	$2/2 = 1$
Fatigue=No	0	1
Travel=No	$2/3$	$1/2$

$P(\text{Pos} | X) = 0$

$P(\text{Neg} | X) > 0$

Prediction: Negative.

Task 5 – Python Implementation

Below is a single, clean implementation that handles all four datasets.

```
from collections import Counter
```

```
def prior(data, target):
```

```
    total = len(data)
```

```
    counts = Counter(row[target] for row in data)
```

```
    return {c: counts[c]/total for c in counts}
```

```
def conditional(data, features, target):
```

```
    cond = {f: {} for f in features}
```

```
    classes = set(row[target] for row in data)
```

```
    for f in features:
```

```
        values = set(row[f] for row in data)
```

```
        for v in values:
```

```
            cond[f][v] = {}
```

```
            for c in classes:
```

```
                subset = [r for r in data if r[target]==c]
```

```
                cond[f][v][c] = sum(r[f]==v for r in subset)/len(subset)
```

```
    return cond
```

```
def predict(priorP, condP, sample):
```

```
    scores={}
```

```
    for c in priorP:
```

```
        p=priorP[c]
```

```
        for f,v in sample.items():
```

```
            p*=condP[f].get(v,{}).get(c,1e-6)
```

```
        scores[c]=p
```

```
    return max(scores,key=scores.get)
```

```
# Example: Dataset 1
```

```
data1=[
```

```
{'Weather':'Rain','Road':'bad','Traffic':'high','Engine':'no','Accident':'yes'},
```

```
{'Weather':'Snow','Road':'average','Traffic':'normal','Engine':'yes','Accident':'yes'},
```

```
{'Weather':'Clear','Road':'bad','Traffic':'light','Engine':'no','Accident':'no'},
```

```
{'Weather':'Clear','Road':'good','Traffic':'light','Engine':'yes','Accident':'yes'},
```

```

{'Weather':'Snow','Road':'good','Traffic':'normal','Engine':'no','Accident':'no'},
{'Weather':'Rain','Road':'average','Traffic':'light','Engine':'no','Accident':'no'},
{'Weather':'Rain','Road':'good','Traffic':'normal','Engine':'no','Accident':'no'},
{'Weather':'Snow','Road':'bad','Traffic':'high','Engine':'no','Accident':'yes'},
{'Weather':'Clear','Road':'good','Traffic':'high','Engine':'yes','Accident':'no'},
{'Weather':'Clear','Road':'bad','Traffic':'high','Engine':'yes','Accident':'yes'}
]
features1=['Weather','Road','Traffic','Engine']
priors1=prior(data1,'Accident')
conds1=conditional(data1,features1,'Accident')
sample1={'Weather':'Rain','Road':'Good','Traffic':'Normal','Engine':'No'}
print("Dataset1 Prediction:",predict(priors1,conds1,sample1))

```

You can reuse the same three functions for Datasets 2–4 by replacing the table, feature names, and target.

Final Predictions Summary

Task	Dataset	Predicted Class
1	Accident	No
2	Play Game	Q1 → No, Q2 → Yes
3	Loan Approval	Q1 → Yes, Q2 → No
4	Disease Diagnosis	Q1 → Positive, Q2 → Negative